Layered Urban Traffic Flow System (LTFS): Modeling and Control Baselines on a Real-City Network

Akinwole Paul-Emmanuel Oreoluwa
School of Control Science and Engineering
Hohai University
Email: akinwolepaulemmanuel@gmail.com

***Abstract*—Urban traffic congestion remains a persistent challenge for large cities, even with decades of progress in adaptive signal control and route guidance. In most networks, short local trips and long through trips share the same road space and signal infrastructure, which leads to recurring queues, spillback, and highly variable travel times. This paper formalizes a Layered Urban Traffic Flow System (LTFS) that separates everyday surface traffic from an express layer intended for longer or more urgent trips, coupled via controllable entry and exit gates.**

Using a Shanghai subnetwork simulated in SUMO, we establish non–reinforcement learning (non-RL) baselines for LTFS. On the surface layer, we implement classical signal controllers: Fixed Time (Webster style), traffic actuated, Max Pressure, and a priority-weighted Max Pressure (PWMP) extension. Within each layer, we consider static shortest path routing, dynamic travel time routing, and distributed Q routing. At inter-layer gates, we introduce two simple gating heuristics: a Time Savings Threshold (TST) policy and an Urgency Weighted Admission (UWA) policy that prioritize vehicles by predicted time saving and urgency while respecting express-layer occupancy constraints.

We define a multi objective LTFS control problem balancing efficiency (delay, travel time, queues), reliability (95th percentile travel time, buffer index), and fairness (class wise inequality), subject to signal, capacity, gate safety, and layer consistency constraints. We then design a simulation methodology and evaluation protocol on a real-city network, including demand scenarios with peak, off-peak, and incident conditions. For benchmarking, wasted time is computed relative to a policy-conditioned free-flow reference obtained under uncongested operation with the same controller logic. The results section describes qualitative trends, while numerical values are placeholders pending full experiment runs. This paper provides a formal LTFS baseline, a clear separation between standard and novel components, and a reproducible evaluation recipe that subsequent RL based work can build on.

***Index Terms*—Urban traffic control, layered networks, traffic signal optimization, max pressure control, priority-weighted max pressure, dynamic routing, Q routing, SUMO, fairness, reliability, Shanghai.**

I. Introduction

A. Motivation

Large metropolitan areas operate road networks that must simultaneously serve short local access trips and long through trips. In practice, these heterogeneous demands share the same surface network and signal infrastructure, which causes recurrent congestion on key arterials, long tail delays for some vehicles, and spillbacks that propagate across multiple intersections.

Conventional adaptive signal systems can improve local efficiency but are limited by the single layer nature of the infrastructure and the fact that all trips compete for the same road space, regardless of urgency. Many cities already have grade separated expressways, elevated corridors, or tunnels, but these are typically designed and operated as separate facilities, without explicit integrated control that coordinates surface signals, routing, and access to the express facilities. This motivates a more unified network view.

B. LTFS concept

We consider a Layered Urban Traffic Flow System (LTFS) with two road layers:

- A *surface layer* handling local access and lower urgency trips at moderate speeds, controlled by traffic signals at intersections.
- An *express layer* providing higher speed, uninterrupted corridors (for example elevated, tunnel, or underground links) primarily targeted at through or higher urgency trips.

The layers are connected by gates, which are controllable entry and exit points. A vehicle can enter the express layer only at designated gates and must leave it at defined exits, after which it continues on the surface network. The key design question is therefore when and for whom to open these gates, and how this interacts with signal control and routing.

C. Problem statement

This work addresses three central questions for LTFS on a real-city network:

1) How should we formally model a two layer urban road system with surface and express layers, gate nodes, urgency aware classes, and basic safety and capacity constraints?
2) What non-RL baseline controllers and routing schemes are appropriate for LTFS, and how do they perform under realistic demand patterns on a real-city subnetwork?
3) How can we evaluate LTFS in terms of efficiency, reliability, and fairness, and clearly distinguish what is standard versus what is new in such a layered design?

We explicitly do not introduce a new RL controller in this paper. Instead, we aim to establish strong baseline strategies and a transparent evaluation framework so that later work, such as hierarchical multi agent RL, can be compared against them.

D. Contributions

The main contributions of this paper are:

1) **LTFS formalization.** We define a network model for a two layer urban road system with surface and express layers, gate nodes, vehicle classes with urgency scores, route-based urgency, and layer consistency and gate safety constraints.
2) **Baseline controllers and routing for LTFS.** We implement Fixed Time (Webster style), traffic actuated, classical Max Pressure, and a priority-weighted Max Pressure (PWMP) extension on the surface layer, together with static shortest path, dynamic travel time routing, and Q routing within each layer, as non-RL baselines for LTFS [1]–[3].
3) **Priority-aware inter-layer gating and signal coupling.** We introduce Time Savings Threshold (TST) and Urgency Weighted Admission (UWA) gate policies with express-layer occupancy caps, and we couple LTFS state to signal control through bounded PWMP weights derived from vehicle-class, layer, and route-based priority.
4) **Evaluation protocol on a real-city subnetwork.** Using a Shanghai subnetwork simulated in SUMO, we specify demand scenarios, metrics, and a simulation workflow that measures efficiency, reliability, robustness, fairness, admission behavior, and express-layer usage. Detailed numerical tables will be added after full simulation runs.
5) **Policy-conditioned free-flow reference.** We define wasted time relative to a policy-conditioned free-flow reference obtained from uncongested simulations under the same controller logic, ensuring fair comparison between baseline MP and LTFS-based policies.
6) **Standard vs new clarification.** We provide a clear inventory of standard building blocks (queue conservation, saturation flow constraints, max pressure equations, dynamic routing, Q routing) versus new LTFS specific formulations (gating rules, occupancy dynamics, PWMP, policy-conditioned free-flow benchmarking, urgency aware utility, and fairness mechanisms), consolidated in Section IV-D and Appendix A.

II. Background and Related Work

A. Urban traffic signal control

Classical fixed time control is often designed using Webster's method, which selects an optimal cycle length and phase splits to minimize average delay at isolated intersections based on critical lane volumes and saturation flow [1]. These methods are widely implemented in practice and remain standard baselines for urban traffic control.

Actuated control uses real time detector measurements (for example occupancy or gap) to extend or terminate green phases within pre specified bounds, improving responsiveness to variable demand. Modern systems such as SCOOT and SCATS implement coordinated variants of such logic across corridors [2].

Max Pressure (MP) control is a more recent decentralized strategy that selects at each junction the phase with the highest pressure, typically defined as an upstream minus downstream queue difference aggregated over movements in that phase. MP has theoretical guarantees of network stability and has shown strong throughput performance in both simulation and field like studies [3], [4].

Recent work has explored RL based signal control, including deep RL, but these are almost always evaluated on single layer networks and assume routing is exogenous [5]. Our aim here is not to replace these methods but to define an LTFS specific baseline environment.

B. Routing and dynamic traffic assignment

Traditional routing in road networks often uses static shortest path algorithms (for example Dijkstra) based on fixed or free flow link travel times [6]. Such assignments cannot adapt to instantaneous congestion and may misrepresent dynamic behavior.

Dynamic Traffic Assignment (DTA) models combine simulation with time dependent shortest paths, where link travel times are periodically updated and new routes are computed, yielding dynamic user optimal or equilibrium solutions [6], [7]. This motivates our baseline dynamic travel time routing, where vehicles re-route based on smoothed travel time estimates.

Q routing, originally proposed for communication networks, maintains per neighbor estimates of expected delivery time and updates them via temporal difference rules similar to Q learning [9]. Q routing has been adapted to various networked systems as a lightweight distributed RL style routing scheme. In this paper we use a tabular Q routing baseline for intra layer routing to connect LTFS to RL style decision making without training a full deep RL controller.

C. Multilayer and grade separated networks

Many cities operate freeway and arterial systems in parallel, often accompanied by ramp metering, high occupancy vehicle lanes, or managed lanes. Within the DTA literature, integrated freeway surface street systems have been modeled via cell transmission based assignment simulation models [8]. However, these works typically do not represent explicit gate nodes with urgency aware admission, or jointly design signal control, routing, and gate policies on a layered urban network.

Our LTFS concept can be seen as urbanizing this idea by embedding express layer access within a dense urban signalized network and tying it explicitly to trip urgency and fairness.

III. System Model and Assumptions

A. Network structure

We model the road network as a directed graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \tag{1}$$

partitioned into two layers:

- Surface layer $S \subset \mathcal{E}$: signalized urban streets.
- Express layer $X \subset \mathcal{E}$: high speed, limited access links (for example elevated or underground).

Nodes $\mathcal{V}$ include signalized intersections on the surface layer, express junctions in the express layer, and gate nodes that connect the layers. Gate nodes are special connectors that allow vehicles to transfer from surface to express (entry gates) or from express back to surface (exit gates). We denote the set of gate nodes by $\mathcal{G}_{\text{gate}}$.

Each link $e \in \mathcal{E}$ has a free flow time τ_e^0, capacity C_e, possibly a static cost function $c_e(x_e)$ such as a Bureau of Public Roads (BPR) type function for static assignments, and time varying observed travel times $\tau_e^{\text{obs}}(t)$ obtained from probes or simulation.

B. Time discretization and queues

We adopt a discrete time model with time step Δt, indexing intervals by $t = 0, 1, 2, \ldots$. At each junction and movement m (for example from an incoming lane to a specific outgoing lane), we maintain a queue length $q_{i,m}(t)$.

Arrivals, departures, and queue evolution follow standard conservation and capacity constraints, including queue update, service bounded by phase and saturation, nonnegativity constraints, and signal compatibility constraints given a phase $\phi(t)$. Exact formulas are recalled in Appendix A.

C. Vehicles, classes, and urgency

Vehicles are generated from origin destination demands with time varying rates. Each vehicle v has an origin, destination, and departure time, and is associated with a SUMO vehicle type (e.g., bus, truck, passenger). In our simulations these type labels are read online via TraCI and are used as the source of vehicle-class priority.

1) Vehicle-class urgency (A): We assign each vehicle a normalized class urgency score $u_v^{\text{class}} \in [0, 1]$ as a deterministic mapping of its type:

$$u_v^{\text{class}} = f_{\text{class}}(c(v)). \tag{2}$$

2) Route-based urgency (C): Let $L_v^{\text{rem}}(t)$ denote remaining route length from the current position to destination along the vehicle's current route. A normalized route urgency is

$$u_v^{\text{route}}(t) = \text{clip}_{[0,1]}\left(\frac{L_v^{\text{rem}}(t)}{L_{\text{ref}}}\right), \tag{3}$$

where $L_{\text{ref}} > 0$ is a reference distance constant.

3) Combined urgency (A+C): We combine class and route urgency as

$$u_v(t) = \lambda\, u_v^{\text{class}} + (1 - \lambda)\, u_v^{\text{route}}(t), \quad \lambda \in [0, 1]. \tag{4}$$

D. Exact LTFS control equations

To make the controller fully reproducible, we write the exact signal-side LTFS equations used to extend classical Max Pressure.

1) Baseline movement pressure: For each movement m at intersection i, the classical movement pressure is

$$\pi_{i,m}(t) = q_{i,m}(t) - \sum_{e \in \text{down}(m)} \rho_{m,e}\, q_e(t), \tag{5}$$

where $q_{i,m}(t)$ is the upstream movement queue, $\rho_{m,e}$ is the turning ratio from movement m to downstream link e, and $q_e(t)$ is the downstream queue.

2) Movement-level urgency aggregation: Let $\mathcal{V}_m(t)$ denote the set of vehicles currently queued on the upstream link of movement m. We define the mean upstream urgency as

$$\overline{u}_{\text{up}}(m, t) = \begin{cases} \frac{1}{|\mathcal{V}_m(t)|} \sum\limits_{v \in \mathcal{V}_m(t)} u_v(t), & |\mathcal{V}_m(t)| > 0, \\ 0, & |\mathcal{V}_m(t)| = 0. \end{cases} \tag{6}$$

3) Layer and gate indicators: We define deterministic movement indicators:

$$\mathbb{I}_{\text{gate}}(m), \qquad \mathbb{I}_{\text{dis}}(m), \qquad \mathbb{I}_{\text{exp}}(m), \tag{7}$$

for gate-feed, discharge, and express-layer movements, respectively.

4) Priority-weight definition: The LTFS-aware movement weight is

$$w_m(t) = \text{clip}_{[1, w_{\max}]}\Big(1 + \alpha\, \mathbb{I}_{\text{gate}}(m)\, \mathbb{I}[o_X(t) \le \kappa] + \beta\, \mathbb{I}_{\text{dis}}(m) + \eta\, \mathbb{I}_{\text{exp}}(m) + \gamma\, \overline{u}_{\text{up}}(m, t)\Big), \tag{8}$$

where $o_X(t)$ is express-layer occupancy, κ is the occupancy cap, and $\alpha, \beta, \eta, \gamma \ge 0$ tune gate-feed, discharge, express, and urgency effects. By construction $w_m(t) \in [1, w_{\max}]$.

5) Priority-weighted movement and phase pressure: The LTFS movement pressure is

$$\pi_{i,m}^{\text{LTFS}}(t) = w_m(t)\, \pi_{i,m}(t), \tag{9}$$

and the corresponding phase pressure is

$$P_i^{\text{LTFS}}(\phi, t) = \sum_{m \in \text{mov}(\phi)} \pi_{i,m}^{\text{LTFS}}(t). \tag{10}$$

The selected phase is

$$\phi_i^\star(t) \in \arg\max_{\phi \in \Phi_i} P_i^{\text{LTFS}}(\phi, t), \tag{11}$$

subject to standard minimum-green and switching constraints.

6) Gate admission rule: For a vehicle v at an LTFS gate, define predicted remaining travel-time saving

$$\Delta T_v(t) = \hat{T}_S(v, t) - \hat{T}_X(v, t). \tag{12}$$

The urgency-weighted admission (UWA) rule is

$$u_v(t)\, \Delta T_v(t) \ge \Theta_u, \qquad o_X(t) \le \kappa, \tag{13}$$

while the TST baseline is recovered by replacing $u_v(t)\, \Delta T_v(t)$ with $\Delta T_v(t)$ and threshold Θ_u with Θ.

At any time, a vehicle is either on the surface layer, on the express layer, or at a gate deciding whether to transfer.

E. Assumptions

We make the following modeling assumptions for this baseline study:

- The express layer is free of internal signals (limited access, higher speed), but its flows are constrained by capacity and safety limits at merges, diverges, and gates.
- The surface layer is controlled by traffic signals at urban intersections.
- Pedestrian flows, lane changes, and detailed car following dynamics are handled by SUMO; we operate at the level of link travel times, queues, and signals for control design.
- Routing and gating policies are non-anticipative beyond their use of smoothed travel times and urgency scores.

IV. LTFS Control Problem Formulation

A. Objective and horizon

We consider a finite time horizon $t = 0, \ldots, T$. The LTFS controller (signals, routing, gating) seeks to minimize a multi objective cost

$$J = w_{\text{eff}} J_{\text{eff}} + w_{\text{rel}} J_{\text{rel}} + w_{\text{fair}} J_{\text{fair}}, \tag{14}$$

where:

- J_{eff} measures efficiency (for example total delay, mean travel time, queue lengths, number of stops).
- J_{rel} measures reliability (for example 95th percentile travel time, buffer index).
- J_{fair} measures fairness across classes (for example Gini coefficient of class wise mean travel times or delays).

The weights $w_{\text{eff}}, w_{\text{rel}}, w_{\text{fair}} \geq 0$ encode operator preferences.

B. Constraints

The system is subject to:

- *Signal constraints*: one phase active per junction at a time; phase compatibility; minimum and maximum green; intergreen and clearance constraints.
- *Capacity constraints*: saturation flow limits on movements, link capacities, and merge or diverge capacities on express links and gates.
- *Layer consistency constraints*: vehicles must enter and exit the express layer only at gate nodes; no teleportation or illegal layer jumps.
- *Safety constraints at gates*: ensuring safe headways and avoiding over saturation of the express layer.

C. Control policy space

We focus on non-RL baseline policies, grouped as:

1) Signal controllers for the surface layer (Section V).
2) Intra layer routing policies for both layers (Section VI).
3) Inter layer gating heuristics at gate nodes (Section VII).

In this paper, we do not solve the exact optimal control problem for J. Instead, we design and simulate heuristic policies that approximate reasonable operating points and allow us to evaluate the LTFS design.

D. Standard versus novel components in LTFS

To make the provenance of our mathematical model explicit, we distinguish standard building blocks imported from the literature from the new LTFS specific components introduced in this work.

Standard components (S1 to S9) include:

- Movement level queue conservation and service bounds by phase and saturation flow.
- The Max Pressure phase score and phase selection rules.
- Exponential smoothing of link travel times for dynamic routing.
- Standard link cost functions (for example BPR type) for static assignment.
- Classical Q routing temporal difference updates.
- Standard definitions of reliability (for example buffer index) and fairness (for example Gini coefficient).

Exact formulas are recopied in Appendix A for completeness. We do not claim these as original; they follow established formulations in traffic control and routing [3], [6], [9].

Novel LTFS components (N1 to N9) introduced here include:

- Time Savings Threshold (TST) gating rule that compares predicted remaining travel times on surface versus express and checks an express layer occupancy cap.
- Urgency Weighted Admission (UWA) rule that multiplies the predicted time saving by an urgency score, enabling differentiated access for classes such as buses or emergency vehicles.
- Explicit gate safety constraints (merge or diverge and headways) integrated into the gating rule.
- An express layer occupancy state updated via gate inflow and outflow dynamics and constrained in $[0, 1]$.
- Optional quota or credit based fairness mechanisms that allow classes to accumulate credits when denied access and spend them when admitted.
- An occupancy penalized gate utility that trades time savings against congestion risk on the express layer.
- A layer consistent path structure requiring routes to respect the Surface to Express to Surface structure with entry and exit only at gates.
- Priority-Weighted Max Pressure (PWMP), a bounded extension of classical Max Pressure in which movement pressures are amplified by vehicle-class, layer, and route-based priority terms while preserving positive, clipped weights.
- A policy-conditioned free-flow reference for wasted-time computation, defined from uncongested simulations under the same policy logic rather than from a single global fixed-time or signal-free reference.

A detailed inventory of S1 to S9 and N1 to N9, along with their equations and examples, is given in Appendix A, which also explains which parts are standard and which are claimed as novel contributions of this LTFS baseline.

V. Surface Layer Control Baselines

We implement three classical signal control strategies and one LTFS-aware extension on the surface layer, which together serve as strong non-RL baselines.

A. Fixed Time (Webster style) control

The fixed time controller uses Webster style design for isolated intersections:

1) Estimate critical lane volumes for each phase and compute saturation flow rates.
2) Use Webster's formula to compute an approximate optimal cycle length that minimizes average delay under given volumes [1].
3) Allocate green splits to phases in proportion to critical flows, adjusted for lost times.
4) For arterials, optionally define offset plans to achieve green waves for dominant directions.

This controller does not adapt to real time fluctuations but provides a robust baseline based on average demand.

B. Actuated control

The actuated controller uses detectors on selected approaches:

- Each phase has minimum green, maximum green, and gap parameters.
- If vehicles are detected within a certain extension window, the green can be extended up to the maximum.
- If no vehicles are detected before the gap expires, the controller terminates the green early and moves to the next phase in the sequence.

In corridor settings, this logic can be embedded in a fixed cycle with offsets to emulate SCOOT or SCATS like progression [2].

C. Max Pressure control

The Max Pressure controller is implemented at each signalized intersection:

1) For each phase ϕ, compute a pressure score as a sum of upstream queues minus weighted downstream queues for the movements served by ϕ.
2) At each decision time, select the phase with maximum pressure, subject to minimum green and switching constraints.

Max Pressure is known to be throughput optimal under certain conditions and tends to perform well in congested networks [3], [4]. This makes it an appropriate strong baseline for LTFS.

D. Priority-Weighted Max Pressure (PWMP) control

To incorporate (A) vehicle-class priority, (B) LTFS layer priority, and (C) route-based priority into signal control while preserving the Max-Pressure structure, we use a bounded movement weight that *amplifies but never suppresses* classical pressure.

For each movement m at intersection i, define classical movement pressure:

$$\pi_{i,m}(t) = q_{i,m}(t) - \sum_{e \in \mathrm{down}(m)} \rho_{m,e}\, q_e(t). \tag{15}$$

The PWMP phase pressure is:

$$P_i^{\mathrm{PWMP}}(\phi, t) = \sum_{m \in \mathrm{mov}(\phi)} w_m(t)\, \pi_{i,m}(t), \tag{16}$$

The selected phase is given by

$$\phi_i^\star(t) \in \arg\max_{\phi \in \Phi_i} P_i^{\mathrm{PWMP}}(\phi, t), \tag{17}$$

subject to standard switching constraints. Here Φ_i denotes the feasible phase set at intersection i, and $q_e(t)$ denotes the queue on downstream link e.

1) Layer-aware movement classification (B): Each movement is deterministically labeled as: (i) *gate-feed* if its upstream link feeds an LTFS entry gate, (ii) *discharge* if it releases vehicles from the express layer back to the surface, and (iii) *express* if its upstream link lies on the express layer.

2) Weight construction (A+B+C): We use:

$$w_m(t) = \mathrm{clip}_{[1, w_{\max}]}\Big(1 + \alpha\, \mathbb{I}_{\mathrm{gate}}(m)\, \mathbb{I}[o_X(t) \le \kappa] + \beta\, \mathbb{I}_{\mathrm{dis}}(m) + \eta\, \mathbb{I}_{\mathrm{exp}}(m) + \gamma\, \overline{u}_{\mathrm{up}}(m, t)\Big), \tag{18}$$

where $o_X(t)$ is express occupancy, κ is the occupancy cap, and $\overline{u}_{\mathrm{up}}(m, t)$ is the mean urgency of vehicles on the upstream link of m, computed from (4). By construction $w_m(t) \ge 1$.

VI. Intra Layer Routing Baselines

We consider three routing paradigms within each layer.

A. Static shortest path routing

In static routing, each vehicle chooses a path at departure based on static link costs, such as free flow times or BPR style generalized costs, and follows this path to completion without rerouting [6]. This is the simplest baseline and represents systems where routing is not dynamically updated.

B. Dynamic travel time routing

For dynamic travel time routing, we maintain smoothed estimates of link travel times:

$$\hat{\tau}_e(t) = \xi \tau_e^{\mathrm{obs}}(t) + (1 - \xi)\hat{\tau}_e(t - 1), \tag{19}$$

with smoothing factor $\xi \in (0, 1]$. At regular intervals (for example every 30 s), we:

1) Update $\hat{\tau}_e(t)$ for all edges based on the latest observations.
2) Compute time dependent shortest paths using these costs.
3) Assign new departures to these dynamic shortest paths; optionally allow mid trip rerouting for long trips.

This approximates a DTA style dynamic user optimal behavior at the level of our simulation [6], [7].

C. Q routing (tabular)

In Q routing, each node i maintains an estimate $Q(i, n, d)$ of the expected remaining travel time to destination d if a vehicle is forwarded to neighbor n. When a vehicle traverses from node i to neighbor n on its way to d, we observe a travel time sample and update:

$$Q(i, n, d) \leftarrow (1-\eta)Q(i, n, d) + \eta \left(t^{\text{obs}}_{i \to n} + \min_{n'} Q(n, n', d) \right), \tag{20}$$

with learning rate ν. For routing decisions, the node chooses the neighbor with minimum $Q(i, n, d)$, occasionally exploring alternatives with probability ε [9].

Q routing is fully distributed and can adapt to congestion without a centralized controller, but may exhibit oscillations if not tuned carefully. Here it serves as a routing baseline that is RL flavored but non deep and easily reproducible.

VII. Inter Layer Gating Strategies (Non RL)

Inter layer gates control which vehicles may enter the express layer and when.

A. Time Savings Threshold (TST) gating

Let $\hat{T}_S(v, t)$ and $\hat{T}_X(v, t)$ be the predicted remaining travel times for vehicle v if it stays on the surface layer versus if it switches to the express layer at time t. Define

$$\Delta T_v(t) = \hat{T}_S(v, t) - \hat{T}_X(v, t). \tag{21}$$

The Time Savings Threshold (TST) gate rule admits v to the express layer if:

1) $\Delta T_v(t) \geq \Theta$, where $\Theta > 0$ is a time savings threshold, and
2) current express layer occupancy $o_X(t)$ is below a capacity cap $\kappa \in (0, 1)$.

If either condition fails, the vehicle remains on the surface path.

B. Urgency Weighted Admission (UWA)

In UWA, urgency is not an external label; it is computed deterministically from (A) vehicle class and (C) route-based urgency using (4). Admission is based on a weighted time saving:

$$u_v(t)\, \Delta T_v(t) \geq \Theta_u, \tag{22}$$

with threshold $\Theta_u > 0$, again subject to the occupancy constraint $o_X(t) \leq \kappa$. This admits high-urgency vehicles under milder conditions than low-urgency vehicles.

C. Gate safety and occupancy dynamics

Gate decisions must obey local safety constraints, including maximum merge and diverge rates, headway constraints on express entry, and physical capacities at gate links. The express layer occupancy $o_X(t)$ is updated based on flows into and out of the express layer and is bounded between 0 and 1. Over admission is prevented by the occupancy cap κ, which can be tuned in sensitivity studies.

D. Optional fairness enhancements

Optionally, we can incorporate quota or credit mechanisms where classes accumulate credit when denied express access and spend it when admitted. This interacts with the fairness term in the objective J and can be activated as an ablation in experiments.

VIII. Methodology and Simulation Design

A. Case study network and simulator

We consider a real-city subnetwork of Shanghai, including a set of arterial corridors, minor streets, and an overlaid express corridor (elevated or underground) with entry and exit gates. The network is implemented in SUMO, with microscopic car following and lane changing models, signal controllers implemented via TraCI, and per vehicle logging of trajectories and travel times. The exact network geometry, demand data, and configuration scripts are prepared for release alongside the paper.

B. Demand scenarios

We define several demand scenarios:

- S0 (single layer baseline): surface network only, no express layer; demand patterns for morning peak, midday, and evening peak.
- S1 (LTFS baseline): layered network with gates active under the selected gating policy.
- S2 (incident variant): LTFS or single-layer runs with a local blockage or lane drop.
- S3 (demand-surge variant): LTFS or single-layer runs with elevated peak demand.

Across these scenarios, the surface layer controller may be fixed time, actuated, classical Max Pressure, or PWMP, and intra-layer routing may be static, dynamic travel time, or Q routing.

C. Controllers and routing configurations

For each scenario, we combine:

- One surface layer controller (fixed time, actuated, classical Max Pressure, or priority-weighted Max Pressure).
- One intra layer routing scheme (static, dynamic travel time, or Q routing).
- One gating strategy (TST, UWA, or no gating as a control).

These combinations yield a grid of baseline configurations that allows us to disentangle the effect of layering, routing adaptivity, gating policy, and priority-aware signal coupling. In the PWMP variants, urgency and layer-coupling terms may be activated or deactivated in ablation form, allowing us to isolate the contribution of vehicle-class priority, route-based urgency, and express-layer signal coupling.

D. Ablation Study Design

To isolate the impact of (A) vehicle-class priority, (B) layer-aware coupling, and (C) route-based priority, we evaluate the ablation suite in Table I. All variants are deterministic and non-learning; only the presence or absence of priority terms and layer coupling differs.

TABLE I
ABLATION SUITE ALIGNED WITH THE PRIORITY-WEIGHTED LTFS EQUATIONS.

ID	LTFS Layers	Gate Rule	Urgency u_v	Signal	$\mathbb{I}_{\text{gate}}$	$\mathbb{I}_{\text{dis}}$	$\mathbb{I}_{\text{exp}}$	$\overline{u}_{\text{up}}$
A0	Surface only	None	N/A	MP	–	–	–	–
A1	Surface+Express	None (open)	N/A	MP	–	–	–	–
A2	Surface+Express	TST	None	MP	–	–	–	–
A3	Surface+Express	UWA	Class only (u^{class})	MP	–	–	–	–
A4	Surface+Express	UWA	Route only (u^{route})	MP	–	–	–	–
A5	Surface+Express	UWA	Class+Route (u)	MP	–	–	–	–
A6	Surface+Express	UWA	Class+Route (u)	PWMP	✓	✓	✓	–
A7	Surface+Express	UWA	Class+Route (u)	PWMP	✓	✓	✓	✓
A8	Surface+Express	UWA	Class+Route (u)	PWMP	✓†	✓	✓	✓

† Gate-feed amplification is occupancy-guarded: $\mathbb{I}_{\text{gate}}(m)\mathbb{I}[o_X(t) \leq \kappa]$ as in (18).

E. Experimental protocol

A typical experiment run proceeds as follows:

1) *Initialization*: warm up period to load queues and stabilize flows (for example 900 s). Initialize Q routing tables if used.
2) *Simulation period*: simulate for a fixed duration (for example 5400 s) under the chosen scenario. Apply the selected signal controller, routing, and gating rules at each step.
3) *Logging*: record per vehicle travel time, delays, number of stops, and layer usage. Record link level queues, flows, and gate occupancy over time.
4) *Repeat over seeds*: perform multiple runs with different random seeds for demand, initial conditions, or stochastic elements.

F. Validation strategy

We do not re-prove the known mathematical properties of queue conservation, saturation constraints, or Max Pressure stability. Instead, simulations are used to validate the LTFS specific design choices, including whether the two layer structure plus gating yields improved efficiency and reliability compared to a single layer baseline, how sensitive performance is to thresholds Θ, Θ_u, and κ, how fairness across classes changes under different gating policies, and how robust the LTFS baselines are to incidents and demand surges.

A detailed inventory of the formulas and their provenance is provided in Appendix A, so that readers can see precisely which equations are standard and which are new.

IX. EVALUATION METRICS

A. Free-Flow Reference and Wasted Time Definition

Wasted time (WT) is defined as the excess travel time relative to a free-flow reference:

$$\mathrm{WT}^{\pi} = \mathrm{TT}^{\pi} - \mathrm{TT}^{\pi}_{\text{ff}}, \tag{23}$$

where TT^{π} is the realized travel time under control policy π (baseline Max Pressure or an LTFS policy) and $\mathrm{TT}^{\pi}_{\text{ff}}$ is a lower-bound reference for the *same* route and vehicle class under that same policy.

a) Why a policy-conditioned free-flow reference (Option 1A).: In signalized urban networks, a valid free-flow reference must satisfy three conditions: (i) it must be achievable under the same right-of-way constraints (traffic signals and conflict rules), (ii) it must retain *unavoidable* signal delay that persists even without congestion, and (iii) it must respect policy-induced feasibility constraints (e.g., LTFS gating and layer access). If a single global reference is reused across policies while policies change the admissible trajectories, WT can become biased or even negative for a large fraction of trips, which undermines its interpretation as *congestion-induced* inefficiency.

Accordingly, we compute $\mathrm{TT}^{\pi}_{\text{ff}}$ from an *uncongested simulation executed under the same policy* π. Practically, we run the same network with the same traffic-light programs and the same controller logic, but remove congestion by spreading departures in time (departure stretching) so that interaction effects are negligible while signal timing and any LTFS gate rules remain active. For each route–vehicle class pair, the policy-conditioned free-flow reference is the minimum observed travel time:

$$\mathrm{TT}^{\pi}_{\text{ff}}(r, c) = \min \left\{ \mathrm{TT}^{\pi}_{\text{low}}(r, c) \right\}, \tag{24}$$

where $\mathrm{TT}^{\pi}_{\text{low}}$ denotes travel times observed in the low-demand (or departure-stretched) run under policy π.

This choice makes $\mathrm{TT}^{\pi}_{\text{ff}}$ a policy-consistent lower bound that includes unavoidable signal delay and any unavoidable policy effects (e.g., a gate that remains closed under its own rules even when demand is low), but excludes congestion-induced queueing delay. Therefore, WT^{π} measures *excess delay above the best achievable performance of the same policy* and is comparable across policies as a normalized "gap-to-best" metric. To avoid any ambiguity, we report both absolute TT (system efficiency) and WT (excess above policy-consistent free flow).

We do *not* use alternatives such as disabling traffic signals, computing travel time from speed limits, or using a free-flow reference from a different policy, because these approaches do not preserve signal physics and can either overestimate or underestimate the achievable lower bound, breaking the intended interpretation of (23).

In addition to aggregate travel time (TT) and wasted time (WT), we report (i) class-wise TT/WT by SUMO vehicle

type, (ii) admission rate at gates, (iii) express usage (distance or time on express edges), and (iv) occupancy exceedance statistics (fraction of time $o_X(t) > \kappa$), to verify that priority mechanisms produce intended effects without suppressing low-priority flows.

We compute metrics along four dimensions.

B. Efficiency

Efficiency metrics include mean travel time across all trips, total delay relative to free flow travel time, mean and maximum queue lengths, and throughput measured as vehicles served per hour.

C. Reliability

Reliability metrics include the 95th percentile travel time T_{95} and the buffer index

$$\text{BI} = \frac{T_{95} - T_{50}}{T_{50}}, \tag{25}$$

where T_{50} is the median travel time.

D. Fairness

Fairness is quantified via the Gini coefficient over class wise mean delays or travel times. Lower values indicate more equitable performance across classes, such as between private cars and buses.

E. Robustness

Robustness is assessed by performance degradation under incident scenarios and demand surges, compared to base conditions, across the above metrics.

X. Results and Discussion

At the time of writing, full numerical experiments are still in progress. The trends described here are based on partial runs and design expectations and should be treated as qualitative; concrete numbers will be updated once all simulations are completed.

The final presentation of results will follow the reporting structure in Section X-E, so that each numerical claim is aligned with the ablation design and the policy-conditioned free-flow metric definition.

In addition to comparing layered and non-layered traffic management, we explicitly compare classical Max Pressure against priority-weighted Max Pressure (PWMP) to isolate the effect of priority-aware signal coupling. All wasted-time results are computed using policy-conditioned free-flow references, so WT is interpreted as controller-relative excess delay rather than deviation from a single global free-flow ideal.

A. Overall impact of layering and gating

Preliminary simulations comparing the single layer baseline (S0) to layered LTFS scenarios (S1 to S3) suggest that introducing an express layer with proper gating can reduce mean travel time during peak periods, especially for longer trips. Vehicles that make longer cross network trips benefit most, as they can bypass several signals using the express layer. Short local trips may see modest improvements or near neutral impact, since they were less affected by severe congestion in the baseline.

The most visible benefit appears in tail behavior: the 95th percentile travel time decreases in LTFS scenarios, indicating fewer extreme delays for worst case trips.

B. Influence of signal control

Comparing surface layer controllers under LTFS:

- Fixed time provides a reasonable baseline but cannot adapt to incident induced congestion.
- Actuated control improves performance in moderate and variable conditions, especially when detector placement and gap settings are tuned to local flows.
- Classical Max Pressure tends to perform best among the standard signal controllers in heavily congested or incident scenarios, as it aggressively serves the most overloaded movements and helps prevent spillback.
- PWMP extends this behavior by biasing phase selection toward gate-feed, discharge, express, and high-urgency movements while preserving the Max-Pressure structure; its purpose is not to replace MP conceptually, but to test whether LTFS-aware priority coupling improves layered coordination.

In LTFS, both classical Max Pressure and PWMP are especially relevant for ensuring that vehicles emerging from the express layer do not create new bottlenecks at downstream surface intersections, reinforcing their role as strong baselines for future RL comparisons.

C. Influence of routing strategy

With LTFS enabled:

- Static routing can under utilize the express layer or overload certain gates, because it does not respond to changing congestion.
- Dynamic travel time routing balances flows more effectively, especially under peaked demand or partial incidents, by directing vehicles towards less congested gates and corridors.
- Q routing adapts quickly to local conditions and can find effective detours when incidents occur, but may require careful tuning to avoid oscillations.

Overall, dynamic routing appears to offer a good trade off between adaptivity, stability, and implementation simplicity in the LTFS context, while Q routing is attractive as a distributed baseline that bridges towards RL.

D. Influence of gating policy and thresholds

When comparing TST and UWA:

- TST yields straightforward time saving based admission that is easy to interpret and tune.
- UWA introduces class dependent bias that can substantially benefit high urgency classes, such as buses, freight, or emergency vehicles, sometimes at a small cost in average performance for low urgency vehicles.

Placeholder for Shanghai subnetwork map showing surface layer, express layer, and gate locations.

Fig. 1. Case-study network and LTFS layer layout. Surface links, express links, and gate nodes should be visually distinguished.

Placeholder for CDF or violin/box plot of travel times for baseline MP, LTFS+MP, and LTFS+PWMP.

Fig. 2. Travel-time distribution comparison across the main baseline and LTFS variants. This figure should highlight both central tendency and tail behavior.

Placeholder for bar chart of ablation results A0–A8, using mean TT and/or mean WT.

Fig. 3. Ablation summary isolating the effect of (A) vehicle-class priority, (B) layer-aware coupling, and (C) route-based priority.

Placeholder for time series of gate admissions and express-layer occupancy.

Fig. 4. Gate admission dynamics and express-layer occupancy under LTFS. This figure verifies whether the gating policy respects occupancy constraints while utilizing the express layer effectively.

Thresholds Θ, Θ_u, and occupancy cap κ define a trade off frontier between efficiency, reliability, and fairness. By sweeping these parameters, operators can select operating points that favor, for example, more robust travel times for buses at the expense of slightly longer times for low urgency private trips.

E. Result table and figure structure

To ensure the final results section is directly aligned with the method and ablations, we will report the following core table and figures.

1) Main numerical table: Table II is the structure used for the final numerical summary.

2) Figure 1: network and layer layout:

3) Figure 2: travel-time distribution:

4) Figure 3: ablation summary:

5) Figure 4: gate usage and express occupancy:

6) Figure 5: class-wise fairness:

XI. Limitations and Future Work

This paper focuses on non-RL baselines on a single real-city subnetwork and makes several simplifying assumptions:

- The express layer is modeled without internal signals; in reality, expressways may have ramp metering, variable speed limits, or incidents that require more detailed modeling.
- The routing and gating policies are myopic heuristics that use smoothed travel times and urgency scores but do not plan over long horizons.
- Pedestrian flows, bike lanes, and public transport dwell times are not explicitly optimized in the baseline controllers.

Future work will introduce hierarchical multi agent RL that jointly optimizes gate admission and signal control on LTFS, explore different network topologies and case cities, incorporate multi modal flows and environmental metrics, and study coordination strategies that combine Max Pressure with offset based progression for corridors.

XII. Conclusion

We have presented a formal LTFS baseline for a real-city subnetwork, integrating a two layer network model with surface and express layers, classical signal controllers (fixed time, actuated, Max Pressure), a priority-weighted Max Pressure (PWMP) extension, static, dynamic, and Q routing schemes, and urgency-aware gating policies (TST, UWA) with occupancy constraints. We explicitly separate standard building blocks from novel LTFS specific components, provide an evaluation protocol on a Shanghai subnetwork in SUMO, and define policy-conditioned free-flow references for controller-consistent WT evaluation.

This baseline is intended as a reference environment for subsequent work, including RL based controllers, and as a conceptual bridge between traditional urban traffic control and layered, urgency aware network design.

Appendix A
Standard vs Novel Components, Provenance, and Worked Examples

A. Standard building blocks (S1 to S9)

We list the standard formulas used in this paper.

(S1) Queue conservation

$$q_{i,m}(t+1) = q_{i,m}(t) + \lambda_{i,m}(t) - \mu_{i,m}(t), \tag{26}$$

where $q_{i,m}(t)$ is the queue length on movement m at junction i, $\lambda_{i,m}(t)$ arrivals, and $\mu_{i,m}(t)$ departures during time step t.

TABLE II
MAIN RESULTS TEMPLATE FOR FINAL SIMULATION OUTPUTS. REPLACE PLACEHOLDERS WITH FINAL AVERAGES OVER RUNS.

Variant	Mean TT	Median TT	P95 TT	BI	Mean WT	Gini(TT)	Gate Admit.	Express Use
A0 (Surface MP)	–	–	–	–	–	–	–	–
A2 (LTFS+TST+MP)	–	–	–	–	–	–	–	–
A5 (LTFS+UWA+MP)	–	–	–	–	–	–	–	–
A7 (LTFS+UWA+PWMP)	–	–	–	–	–	–	–	–
A8 (Full LTFS, occ.-guarded PWMP)	–	–	–	–	–	–	–	–

Placeholder for class-wise mean TT/WT or fairness plot (cars, buses, trucks, etc.).

Fig. 5. Class-wise travel time or policy-relative excess delay, used to assess priority effects and fairness across vehicle classes.

(S2) Service bounded by phase and saturation

$$\mu_{i,m}(t) \le s_{i,m}(\phi_i(t)), \tag{27}$$

where $s_{i,m}(\phi)$ is the saturation flow offered to movement m under phase ϕ.

(S3) Max Pressure movement and phase score

$$\pi_{i,m}(t) = q_{i,m}(t) - \sum_{e \in \mathrm{down}(m)} \rho_{m,e}\, q_e(t), \tag{28}$$

and the classical phase pressure is

$$P_i^{\mathrm{MP}}(\phi, t) = \sum_{m \in \mathrm{mov}(\phi)} \pi_{i,m}(t), \tag{29}$$

with turning ratios $\rho_{m,e}$ and downstream link queues $q_e(t)$ [3].

(S4) Max Pressure phase selection

$$\phi_i^\star(t) \in \arg\max_{\phi \in \Phi_i} P_i^{\mathrm{MP}}(\phi, t), \tag{30}$$

possibly with constraints on minimum green and phase switching.

(S5) Exponential smoothing of travel times

$$\hat{\tau}_e(t) = \xi \tau_e^{\mathrm{obs}}(t) + (1-\xi)\hat{\tau}_e(t-1), \tag{31}$$

used to compute time dependent link costs for dynamic routing [6].

(S6) Static link cost function

$$c_e(x_e) = \tau_e^0 \left[1 + a(x_e/C_e)^b\right], \tag{32}$$

a BPR type cost used for static shortest path routing [6].

(S7) Q routing update

$$Q(i,n,d) \leftarrow (1-\eta)Q(i,n,d) + \eta\left(t_{i\to n}^{\mathrm{obs}} + \min_{n'} Q(n,n',d)\right), \tag{33}$$

as introduced in Q routing [9].

(S8) Buffer index for reliability

$$\mathrm{BI} = \frac{T_{95} - T_{50}}{T_{50}}, \tag{34}$$

where T_{95} and T_{50} are the 95th and 50th percentile travel times.

(S9) Fairness via Gini coefficient

The Gini coefficient is computed over class wise mean delays or travel times, using standard formulas.

These items are not claimed as new; they are established in the literature.

B. Novel LTFS specific components (N1 to N9)

The following components are introduced specifically for LTFS in this work.

(N1) Time Savings Threshold (TST) gate admission

$$\Delta T_v(t) = \hat{T}_S(v,t) - \hat{T}_X(v,t), \quad \Delta T_v(t) \ge \Theta, \quad o_X(t) \le \kappa. \tag{35}$$

(N2) Urgency Weighted Admission (UWA)

$$u_v(t)\,\Delta T_v(t) \ge \Theta_u, \quad o_X(t) \le \kappa, \tag{36}$$

where $u_v(t)$ is the combined urgency score defined in (4).

(N3) Gate safety constraints

Merge and diverge capacities and headway constraints are enforced at each gate node, and gate inflows must respect these limits.

(N4) Express layer occupancy dynamics

Express layer occupancy $o_X(t)$ is updated from inflows and outflows and constrained to $[0,1]$.

(N5) Optional quota or credit based fairness mechanism

Classes can accumulate credits when denied gate access and redeem them later, affecting admission decisions.

(N6) Occupancy penalized gate utility

$$U(g,v,t) = u_v \Delta T_v(t) - \lambda_o \psi(o_X(t)), \tag{37}$$

with increasing penalty function ψ and weight λ_o. Admission occurs when $U(g,v,t) \ge 0$.

(N7) Layer consistent path structure

Feasible routes must respect the Surface to Express to Surface pattern; express use is only possible via gates, with no illegal re entry.

(N8) Priority-Weighted Max Pressure (PWMP).

For movement m at intersection i, classical movement pressure is

$$\pi_{i,m}(t) = q_{i,m}(t) - \sum_{e \in \mathrm{down}(m)} \rho_{m,e} q_e(t).$$

The bounded movement weight is

$$w_m(t) = \mathrm{clip}_{[1,w_{\max}]}(1 + \alpha\,\mathbb{I}_{\mathrm{gate}}(m)\mathbb{I}[o_X(t) \le \kappa] + \beta\,\mathbb{I}_{\mathrm{dis}}(m) + \eta\,\mathbb{I}_{\mathrm{exp}}(m$$

and the PWMP phase pressure is

$$P_i^{\mathrm{PWMP}}(\phi, t) = \sum_{m \in \mathrm{mov}(\phi)} w_m(t)\, \pi_{i,m}(t).$$

The selected phase is

$$\phi_i^\star(t) \in \arg\max_{\phi \in \Phi_i} P_i^{\mathrm{PWMP}}(\phi, t).$$

Here $\overline{u}_{\mathrm{up}}(m, t)$ is the mean urgency of vehicles on the upstream link of movement m, computed from (4).

(N9) Policy-conditioned free-flow reference.

For each controller π and route–class pair (r, c), free-flow travel time is defined as

$$\mathrm{TT}_{\mathrm{ff}}^{\pi}(r, c) = \min\left\{\mathrm{TT}_{\mathrm{low}}^{\pi}(r, c)\right\},$$

where $\mathrm{TT}_{\mathrm{low}}^{\pi}(r, c)$ is observed under uncongested operation with the same controller logic. Wasted time is then

$$\mathrm{WT}^{\pi} = \mathrm{TT}^{\pi} - \mathrm{TT}_{\mathrm{ff}}^{\pi}.$$

C. Provenance of models

For clarity in review, we summarize the provenance as follows.

The queue conservation, saturation flow constraints, shortest path routing, Q routing update, and classical Max Pressure phase selection used in this paper follow standard formulations from the traffic control and routing literature. Our contributions lie in adapting these ingredients to a road-to-road layered LTFS architecture with explicit gate nodes and safety constraints, introducing urgency-aware gate admission rules, a priority-weighted Max Pressure (PWMP) extension, and policy-conditioned free-flow references for controller-consistent WT evaluation, and evaluating them with a multi objective efficiency, reliability, and fairness scorecard on a real-city subnetwork in SUMO. We implement this LTFS baseline and prepare scenarios, code, and metrics so that other researchers can replicate and extend layered network control experiments.

D. Worked examples

1) Example 1: Max Pressure phase selection: Consider a junction with two phases:

- Phase 1 serves a movement whose upstream queue is 12 vehicles and whose downstream queue is 5 vehicles.
- Phase 2 serves another movement with upstream queue 6 and downstream queue 9.

Using the simplified movement-pressure formula $\pi = q_{\mathrm{up}} - q_{\mathrm{down}}$:

$$\pi_1 = 12 - 5 = 7, \quad \pi_2 = 6 - 9 = -3. \tag{38}$$

The controller selects the phase serving Movement 1, since it has higher pressure and is expected to produce a larger reduction in overall congestion.

2) Example 2: Gate admission decision under TST and UWA: Suppose at a gate we predict:

- Remaining surface travel time $\hat{T}_S = 18$ minutes,
- Remaining express travel time $\hat{T}_X = 12$ minutes,

so that $\Delta T = 6$ minutes.

With TST:

- If $\Theta = 5$ minutes and express occupancy $o_X < \kappa$, the vehicle is admitted.

With UWA:

- If urgency $u_v = 0.5$, then $u_v \Delta T = 3$ minutes; with $\Theta_u = 4$ minutes, the vehicle is not admitted.
- If urgency $u_v = 1.0$, then $u_v \Delta T = 6$ minutes, which is at least Θ_u, so the vehicle is admitted, assuming occupancy allows.

These examples illustrate how the gate rules operate in concrete terms and how urgency affects gate decisions.

3) Example 3: Priority-Weighted Max Pressure: Consider two candidate phases at an intersection. Suppose Phase 1 serves a gate-feed movement with classical pressure 6, and Phase 2 serves a standard surface movement with classical pressure 7. Under classical Max Pressure, Phase 2 is selected because $7 > 6$. Under PWMP, let the Phase 1 movement have weight $w = 1.4$ due to gate-feed status and elevated upstream urgency, while Phase 2 retains $w = 1.0$. Then the weighted pressures become

$$P_1^{\mathrm{PWMP}} = 1.4 \times 6 = 8.4, \qquad P_2^{\mathrm{PWMP}} = 1.0 \times 7 = 7.0.$$

PWMP therefore selects Phase 1, illustrating how LTFS-aware priority can change phase choice without suppressing classical pressure terms.

References

[1] F. V. Webster, *Traffic Signal Settings*. Road Research Technical Paper No. 39, London, U.K.: Her Majesty's Stationery Office, 1958.

[2] P. Hunt, D. Robertson, R. Bretherton, and M. Royle, "The SCOOT on-line traffic signal optimisation technique," *Traffic Engineering & Control*, vol. 23, no. 4, pp. 190–192, 1982.

[3] P. Varaiya, "The max-pressure controller for arbitrary networks of signalized intersections," in *Advances in Dynamic Network Modeling in Complex Transportation Systems*, New York, NY, USA: Springer, 2013, pp. 27–66.

[4] J. Gregoire, E. Frazzoli, A. de La Fortelle, and T. Wongpiromsarn, "Back-pressure traffic signal control with unknown routing rates," in *IFAC Proceedings Volumes*, vol. 47, no. 3, 2014, pp. 11332–11337.

[5] T. Chu, J. Wang, L. Codecà, and Z. Li, "Multi-agent deep reinforcement learning for large-scale traffic signal control," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 3, pp. 1086–1095, 2020.

[6] Y. Sheffi, *Urban Transportation Networks: Equilibrium Analysis with Mathematical Programming Methods*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1985.

[7] H. S. Mahmassani, "Dynamic network traffic assignment and simulation methodology for advanced system management applications," *Networks and Spatial Economics*, vol. 1, no. 3–4, pp. 267–292, 2001.

[8] H. K. Lo and W. Y. Szeto, "A cell-based variational inequality formulation of the dynamic user optimal assignment problem," *Transportation Research Part B: Methodological*, vol. 36, no. 5, pp. 421–443, 2002.

[9] J. A. Boyan and M. L. Littman, "Packet routing in dynamically changing networks: A reinforcement learning approach," in *Advances in Neural Information Processing Systems 6*, 1994, pp. 671–678.